Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization

The Strategic Product Placement Analysis is a data-driven project focused on understanding how the specific location of a product on a retail shelf—such as eye-level versus bottom-shelf—directly affects its sales performance. By using Tableau Visualization, the project "unveils" or makes visible the hidden relationship between physical placement and consumer purchasing power.

1.Data Collection and Extraction of Data:

Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes, and generate insights from the data.

A) Collect the dataset:

Please use the link to download the dataset:

https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales

B) Understanding the Data

Product ID:

Unique identifier for every item in the store.

Position:

Specific locations like Endcap, Aisle, or Shelf.

Price:

Current selling price compared against Competitor Prices.

Promotion:

Tracks discounts to isolate the impact of placement.

Foot Traffic:

The volume of shoppers passing a specific display.

Demographics:

Target groups like Families, Seniors, or Students.

Category:

Broad groupings such as electronics or groceries.

Seasonality:

Identifies if sales spikes are due to time of year.

Sales Volume: Total units sold to measure "Shelf-Level Lift".

Revenue:

Total income used to calculate ROI per Square Foot.

C) Connecting the dataset with Tableau:

To visualize the dataset in Tableau, import the dataset file into Tableau Desktop. Then, link the relevant columns to Tableau's data fields to create visualizations and analyze the data effectively.

ReferenceVideo:
https://drive.google.com/file/d/1cS7Ork8XG7c_RjdmMW_EwZqQj6cwgn9x/view?usp=sharing

2.Data Preparation:

Data Cleaning: Removing inaccuracies or omissions to ensure the final Tableau analysis is reliable.

Transformation: Formatting and structuring raw variables like Product ID and Sales Volume for software compatibility.

Pattern Discovery: Delving into the dataset to uncover trends between Product Position and consumer behavior.

Relevance Filtering: Refining the scope to focus only on data that addresses your **core objective:** unveiling sales impact.

Targeted Analysis: Using specific subsets, such as Demographics or Seasonal tags, to answer precise business questions.

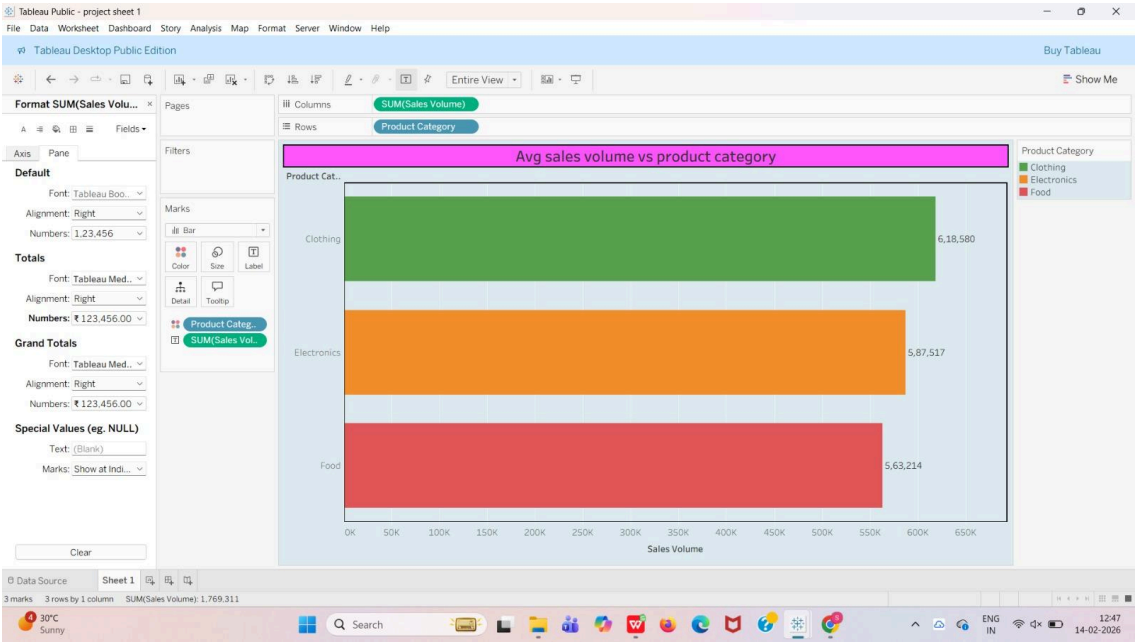
Software Integration: Preparing the finalized dataset for seamless connection with Tableau and Flask.

3.Data Visualization

No of Unique Visualizations

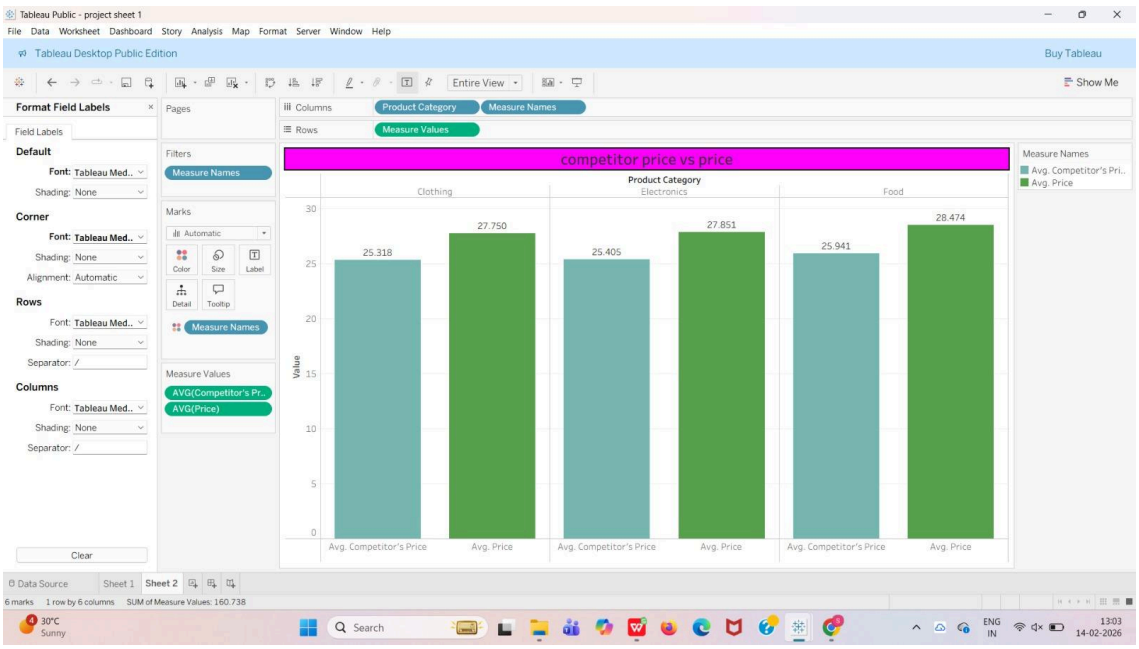
Activity 1.1: Avg Sales Volume vs Product Category

- ➔ Drag Sales Volume to the Columns shelf.
- ➔ Drag Product Category to the Rows shelf.
- ➔ Drag Product Category again to the Columns shelf if needed for comparison.
- ➔ Drag Sales Volume to the Label on the Marks card.
- ➔ Change Sales Volume to Average (AVG).



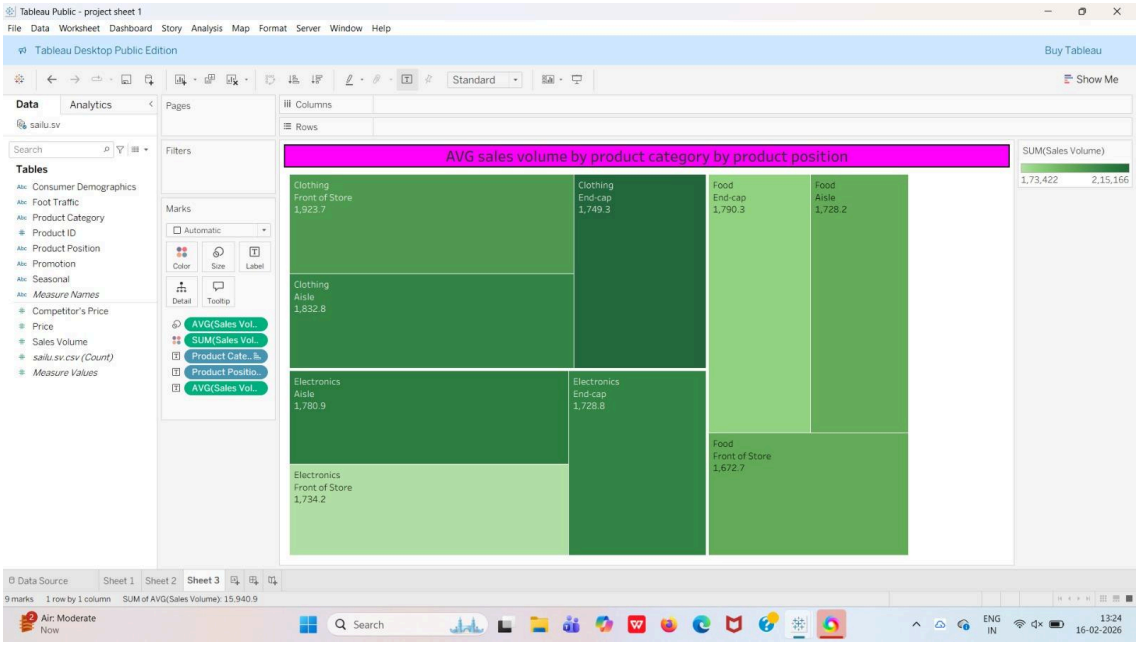
Activity 1.2: Competitor Price Vs Price

- ➔ Drag Product Category and Measure Names to the Columns shelf.
- ➔ Drag Measure Values to the Rows shelf.
- ➔ Drag Measure Names to the Filters shelf and select:
- ➔ Competitor Price
- ➔ Price
- ➔ Drag Measure Names to Color.
- ➔ Set Competitor Price aggregation to Average (AVG).
- ➔ Set Price aggregation to Average (AVG).



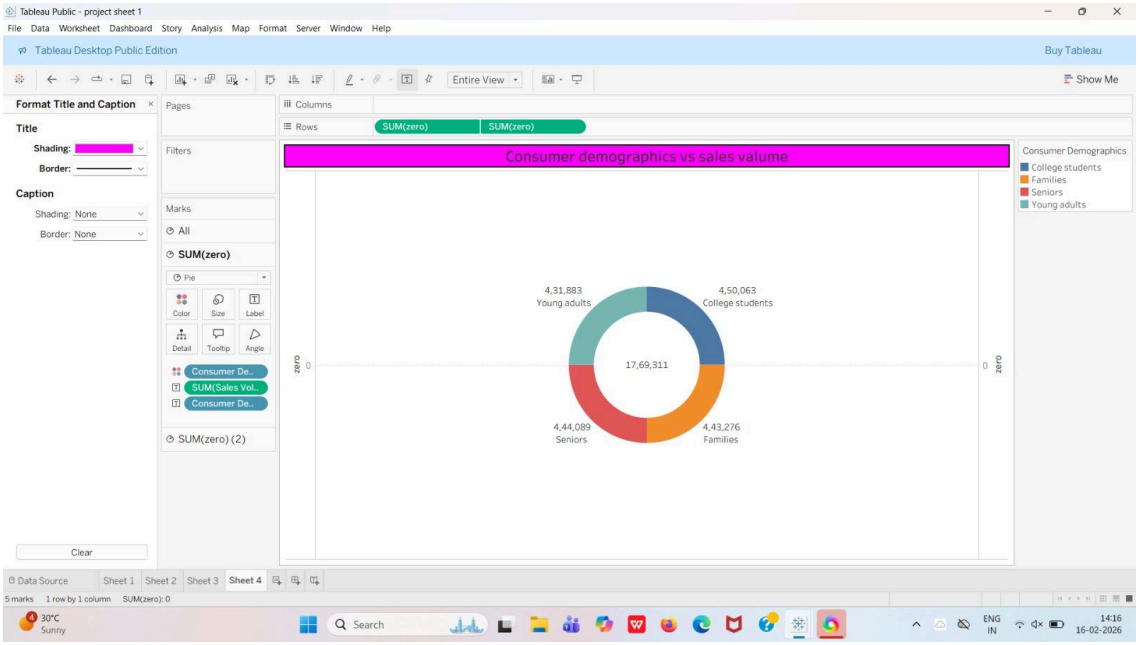
Activity 1.3: Avg Sales Volume by Product Category by Product Position

- ➔ Drag Sales Volume to Size and change aggregation to Average (AVG).
- ➔ Drag Sales Volume to Color.
- ➔ Drag Product Category to Label.
- ➔ Drag Product Position to Label.
- ➔ Drag Sales Volume to Label and set it to Average (AVG).



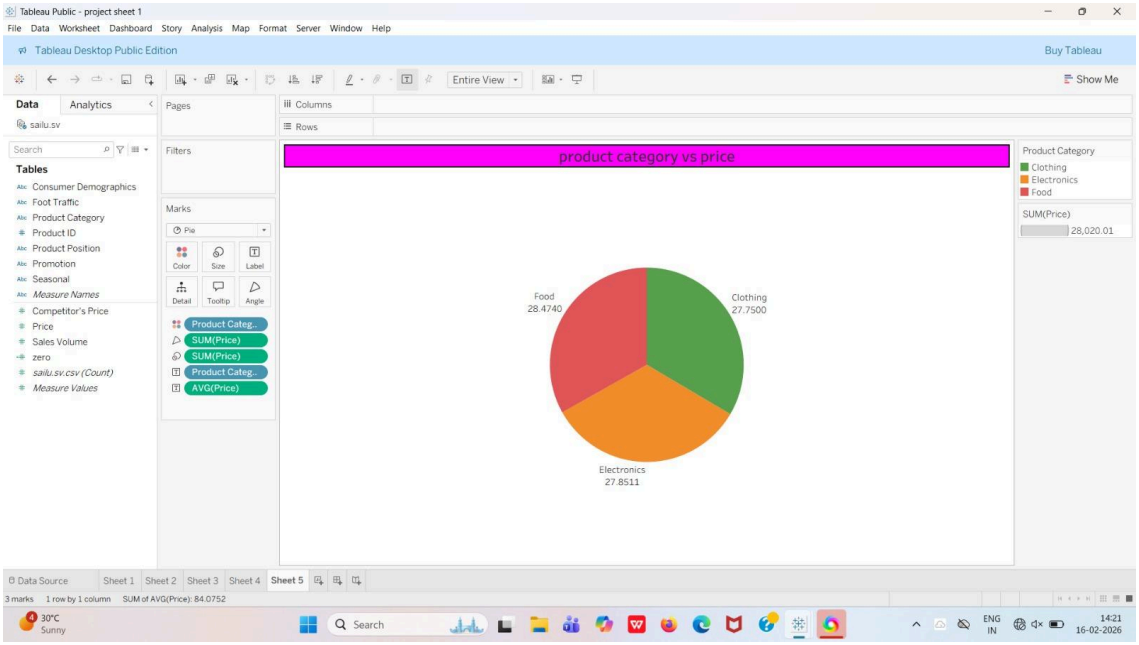
Activity 1.4: Consumer Demographics vs Sales Volume

- ➔ Drag Zero to the Rows shelf twice.
- ➔ Right-click and select Dual Axis.
- ➔ Drag Customer Demographics to Color and Label.
- ➔ Drag Sales Volume to Label.
- ➔ Increase the Size and change the Color as needed.



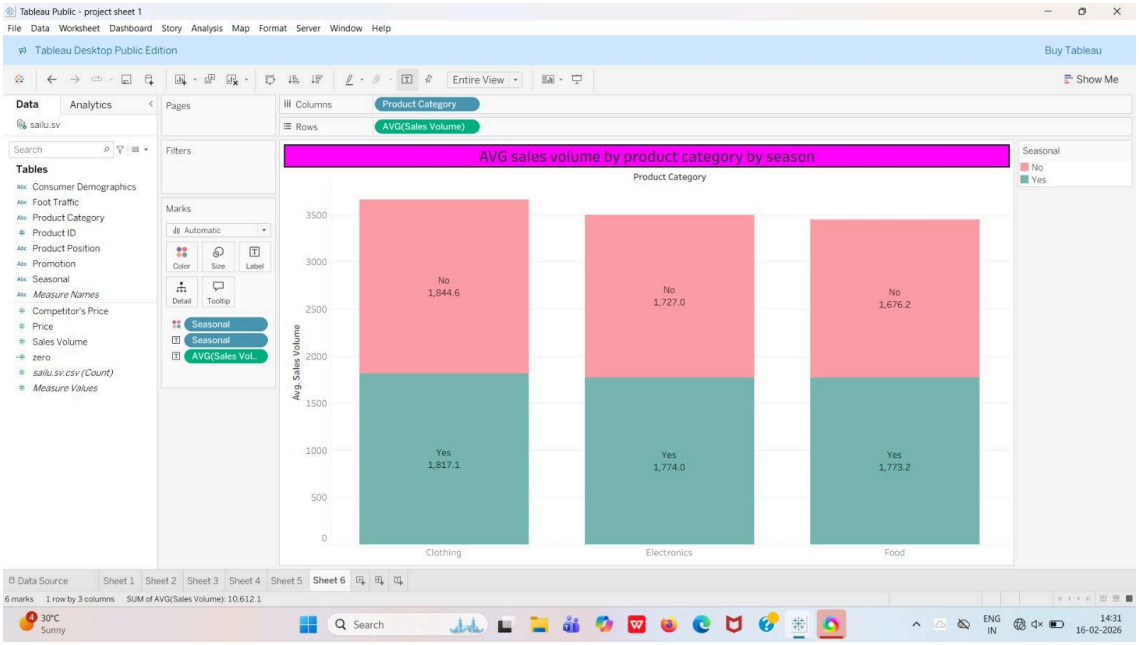
Activity 1.5: Product Category vs Price

- ➔ Drag Product Category to Color.
- ➔ Drag Price to Size.
- ➔ Drag Price to Angle.
- ➔ Drag Product Category to Label.
- ➔ Drag Price to Label.



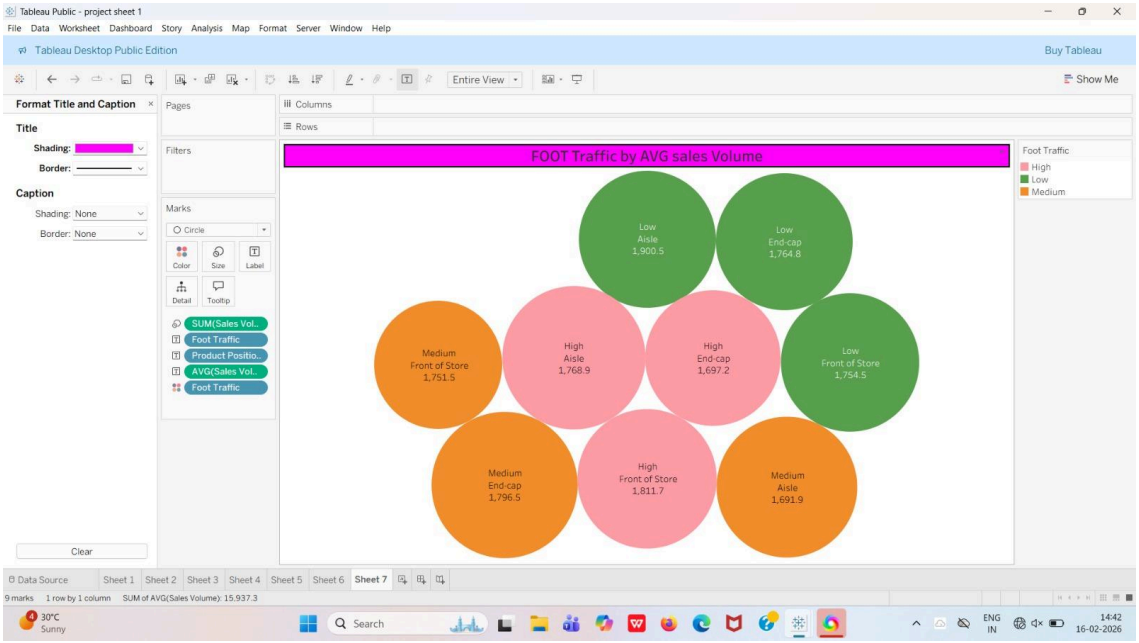
Activity 1.6: Avg Sales Volume by Product Category by Season

- ➔ Drag Product Category to the Columns shelf.
- ➔ Drag Sales Volume to the Rows shelf.
- ➔ Change aggregation to Average (AVG).
- ➔ Drag Season to Color.
- ➔ Drag Season to Label.
- ➔ Drag Sales Volume to Label.



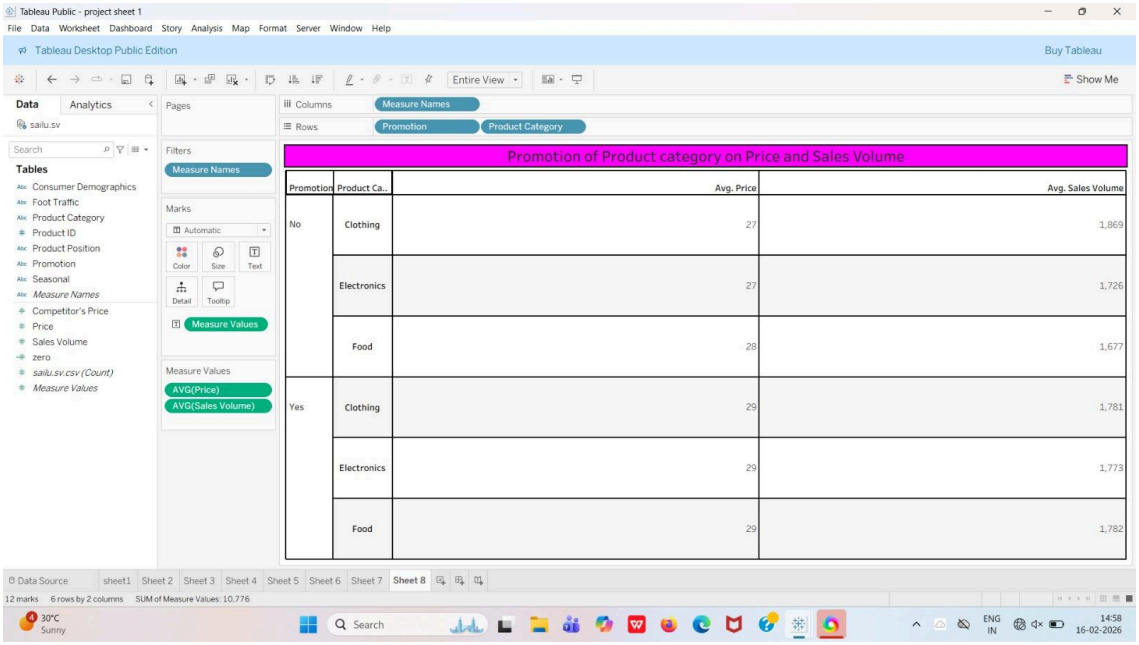
Activity 1.7: Foot Traffic by Avg Sales Volume

- ➔ Drag Sales Volume to Size.
- ➔ Drag Foot Traffic to Label.
- ➔ Drag Product Position to Label.
- ➔ Drag Sales Volume to Label and change aggregation to Average (AVG).
- ➔ Drag Foot Traffic to Color.



Activity 1.8: Promotion of Product Category on Price and Sales Volume

- ➔ Drag Measure Names to the Columns shelf.
- ➔ Drag Promotion to the Rows shelf.
- ➔ Drag Product Category to the Rows shelf.
- ➔ Drag Measure Names to the Filters shelf.
- ➔ Drag Measure Values to Label.
- ➔ Select Average Price and Average Sales Volume.



Promotion	Product Ca..	Avg. Price	Avg. Sales Volume
No	Clothing	27	1,869
	Electronics	27	1,726
	Food	28	1,677
Yes	Clothing	29	1,781
	Electronics	29	1,773
	Food	29	1,782

4) Dashboard:

Responsive and Design of Dashboard:

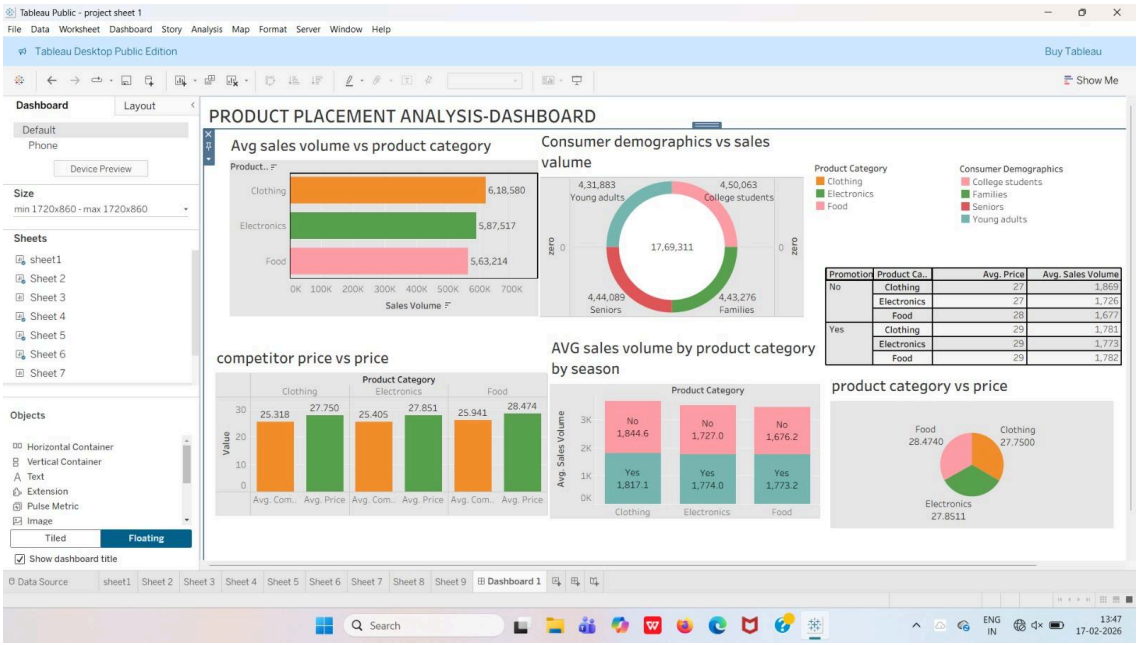
The responsiveness and design of a dashboard for analyzing the performance and efficiency of Product Placement is crucial to ensure that the information is easily understandable and actionable. Key considerations for designing a responsive and effective dashboard include user-centered design, clear and concise information, interactivity, data-driven approach, accessibility, customization, and security. The goal is to create a dashboard that is user-friendly, interactive, and data-driven, providing actionable insights to improve the performance and efficiency of Product Placement Analysis.

Once you have created views on different sheets in Tableau, you can pull them into a dashboard.

Dashboard: Product Placement Analysis – Dashboard

- ➔ Click on Dashboard → New Dashboard.
- ➔ Rename the dashboard as Product Placement Analysis – Dashboard.

- ➔ Select Floating layout.
- ➔ Drag and place each sheet one by one onto the dashboard.
- ➔ Arrange and resize the sheets properly for clear visualization.



5) Story

No of Scenes of Story

The number of scenes in a storyboard for a data visualization analysis of the performance and efficiency of Product Placement will depend on the complexity of the analysis and the specific insights that are trying to be conveyed. A storyboard is a visual representation of the data analysis process and it breaks down the analysis into a series of steps or scenes.

Story: Product Placement Analysis – Story

- ➔ Click on Story → New Story.
- ➔ Rename the story as Product Placement Analysis – Story.
- ➔ Click on the Blank story point.
- ➔ Drag the sheets or dashboard into the story workspace.
- ➔ Add a title and description for each story point to explain the insights.
- ➔ Arrange all sheets in sequence to present the complete analysis story.

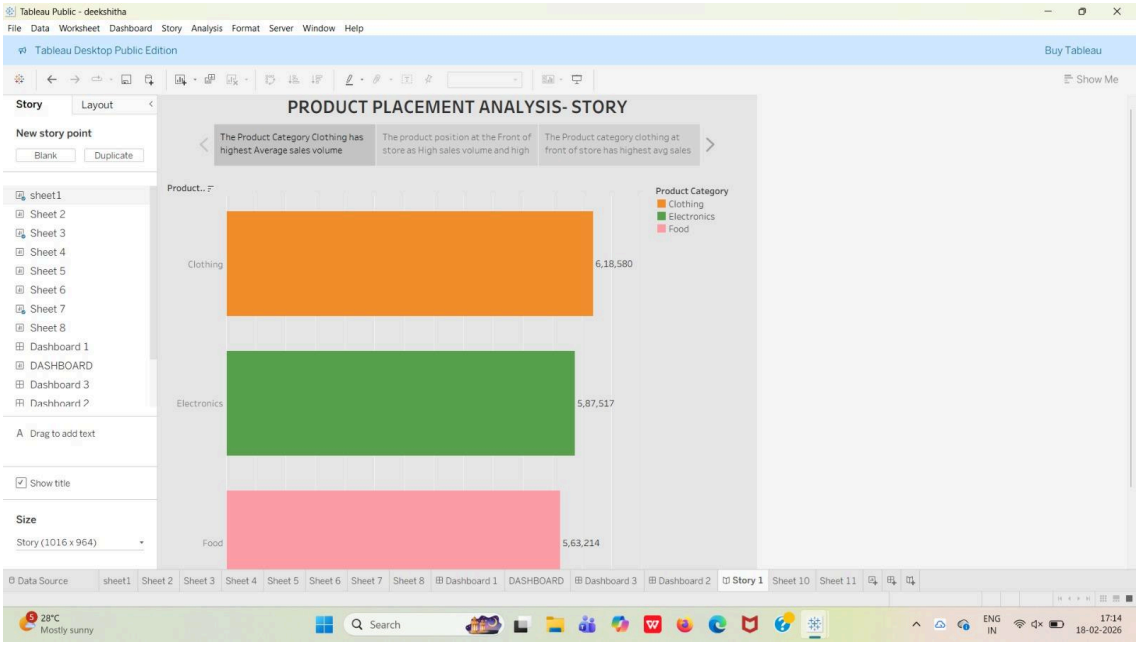


Tableau Public - deekshitha
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Show Me
Story
Layout
New story point
Blank
Duplicate
sheet1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
DASHBOARD
Dashboard 3
Dashboard 2
Drag to add text
Show title
Size
Story (1016 x 964)
PRODUCT PLACEMENT ANALYSIS- STORY
The Product Category Clothing has highest Average sales volume
The product position at the Front of store as High sales volume and high
The Product category clothing at front of store has highest avg sales
Product..
Product Category
Clothing
Electronics
Food
Clothing
6,18,580
Electronics
5,87,517
Food
5,63,214
Data Source
Story 1
Sheet 10
Sheet 11
28°C
Mostly sunny
Search
ENG
IN
17:14
18-02-2026

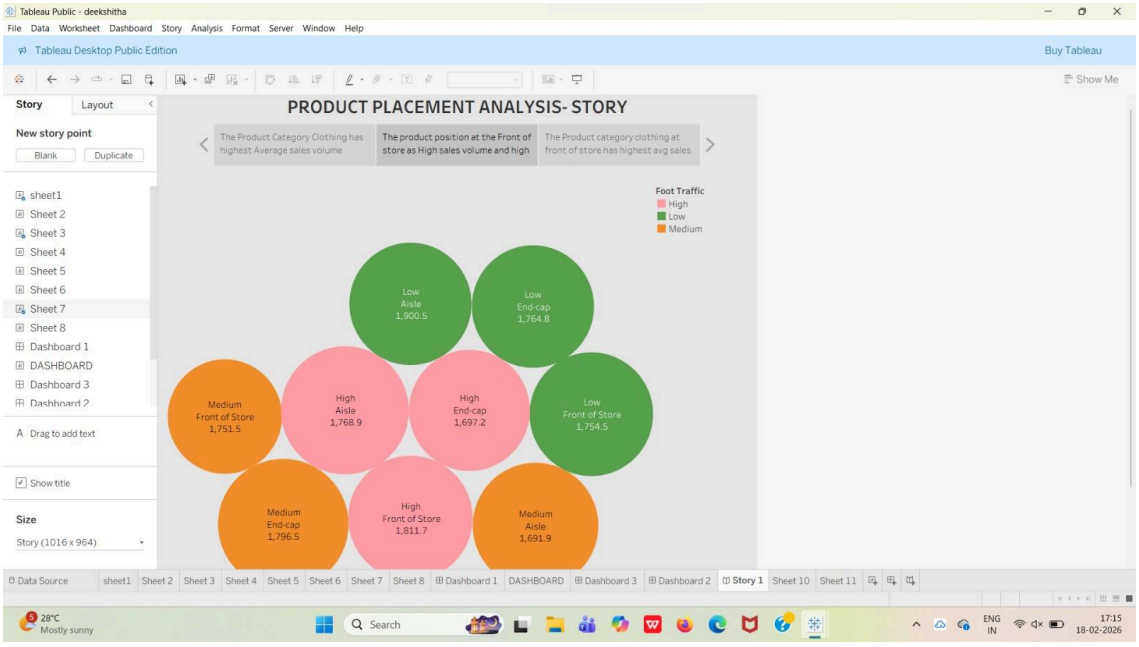
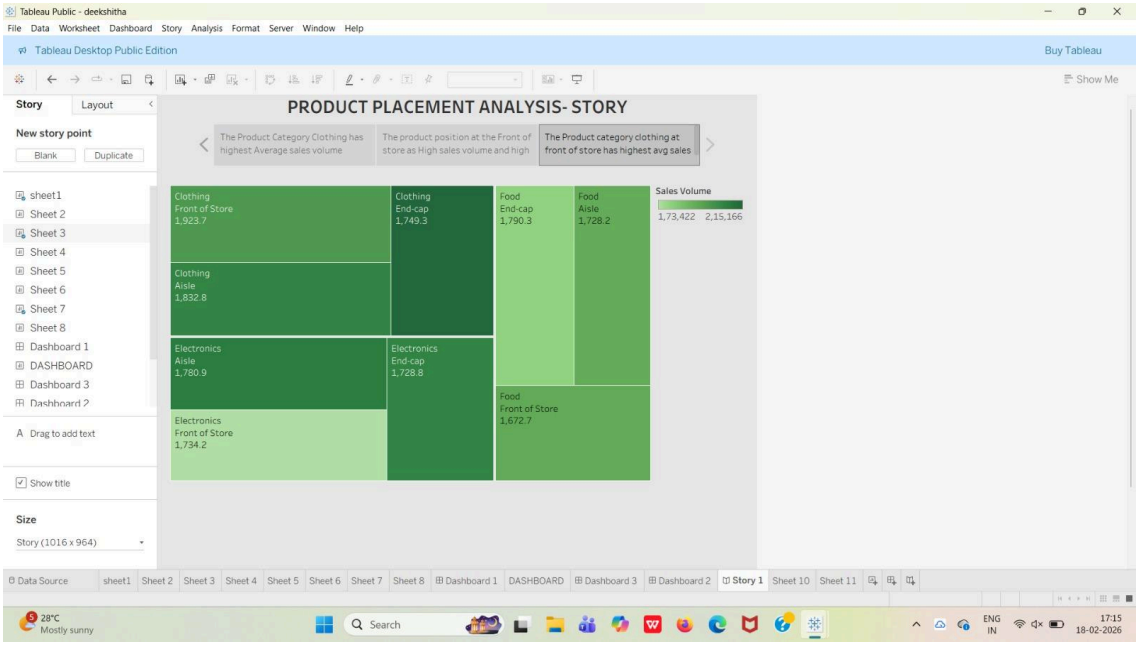


Tableau Public - deekshitha
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Show Me
Story
Layout
New story point
Blank
Duplicate
sheet1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
DASHBOARD
Dashboard 3
Dashboard 2
Drag to add text
Show title
Size
Story (1016 x 964)
PRODUCT PLACEMENT ANALYSIS- STORY
The Product Category Clothing has highest Average sales volume
The product position at the Front of store as High sales volume and high
The Product category clothing at front of store has highest avg sales
Foot Traffic
High
Low
Medium
Low Aisle 1,900.5
Low End-cap 1,764.8
Medium Front of Store 1,751.5
High Aisle 1,768.9
High End-cap 1,697.2
Low Front of Store 1,754.5
Medium End-cap 1,796.5
High Front of Store 1,811.7
Medium Aisle 1,691.9
Data Source
Story 1
Sheet 10
Sheet 11
28°C
Mostly sunny
Search
ENG
IN
17:15
18-02-2026



6)Performance Testing

A) Utilization of Filters

Filters are an indispensable tool in data analysis and visualization, allowing users to refine and focus on specific subsets of data that are relevant to their analysis objectives**.**

B) No of Calculation Fields

Price

Competitors Price

Sales Volume

C) No of Visualizations/ Graphs

Avg Sales Volume by Product Category

Avg Sales Volume by Product Category on Sales

Consumer Demographics vs Product Category

Foot traffic vs Sales Volume

Competitors Price vs Price.

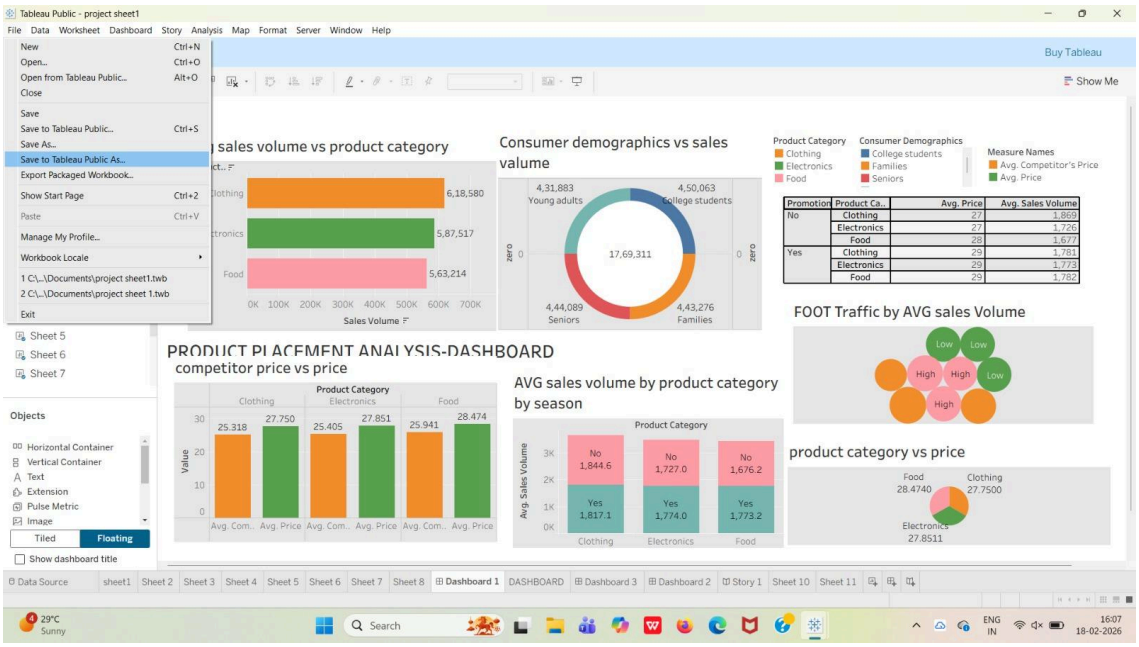
Promotion of Product Category on Price and Sales Volume.

Product Category vs Price

Avg Sales Volume by Product Category by Product Position.

7)Web integration

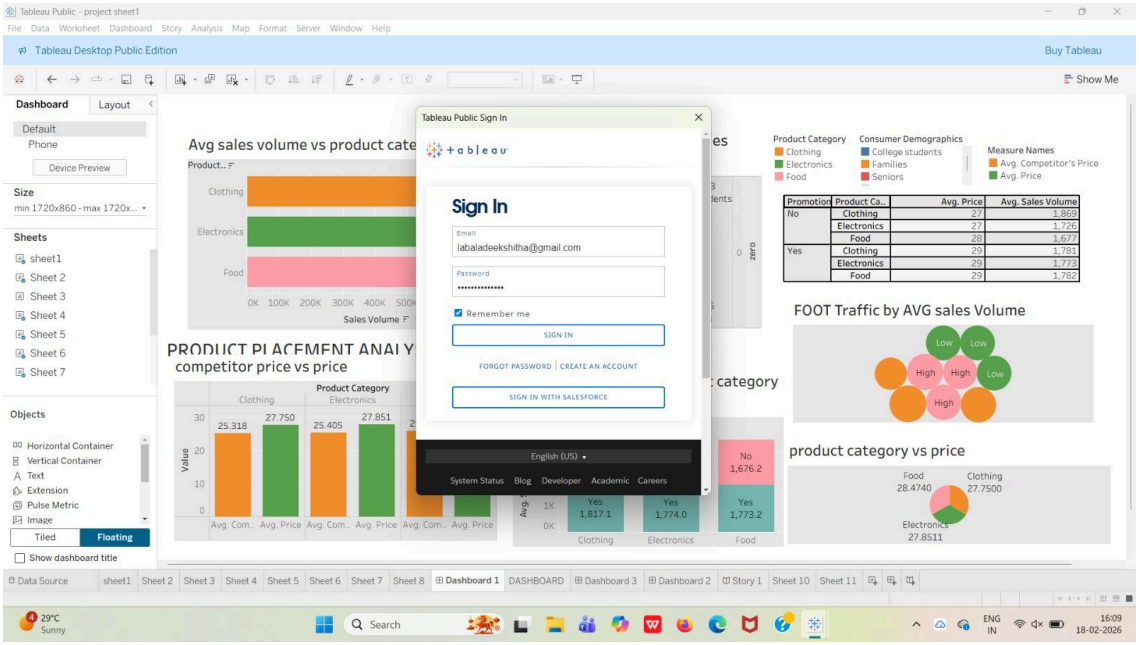
Publishing helps us to track and monitor key performance metrics, to communicate results and progress. help a publisher stay informed, make better decisions, and communicate their performance to others.



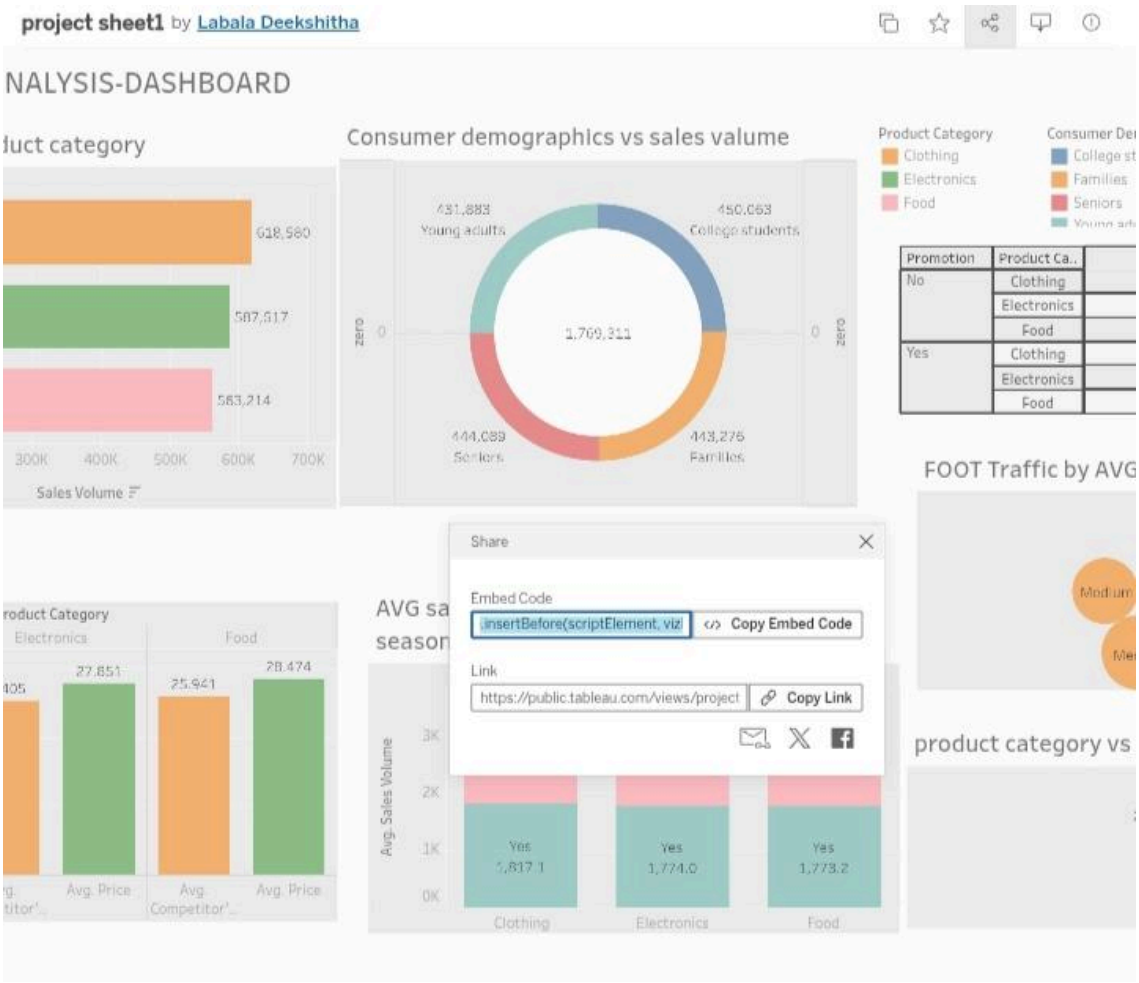
The above gives information on how to save and publish the dashboard to the tableau public.

Click on "File" which is displayed on the left top corner of the dashboard sheet.

Now click on the Save as Tableau Public As option and that will redirect to your sign-in account as shown below.

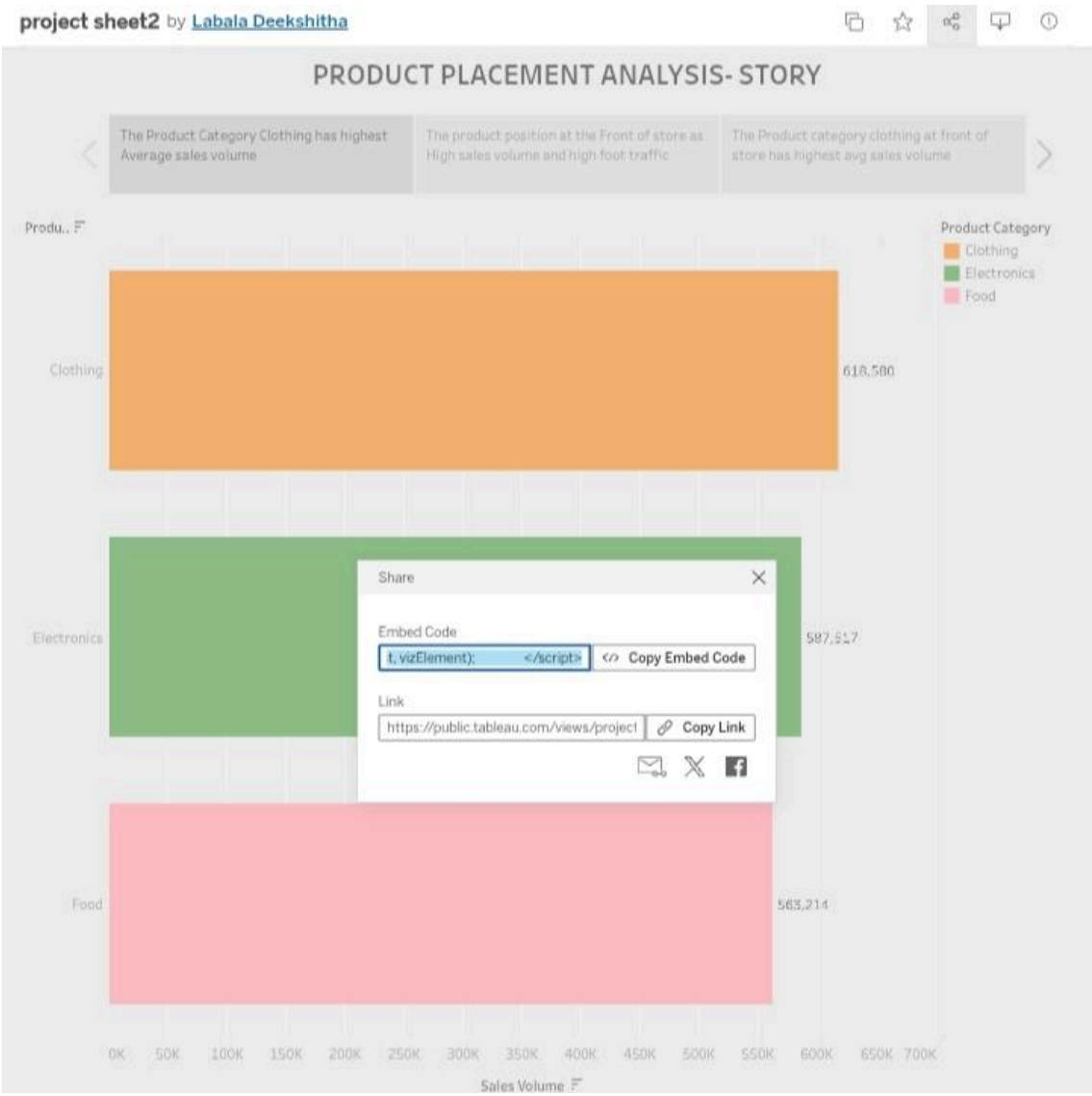


Now sign in to your Tableau public account and there your visualization, dashboard, and story are published. In this way, we can publish your dashboard and story into your tableau public.



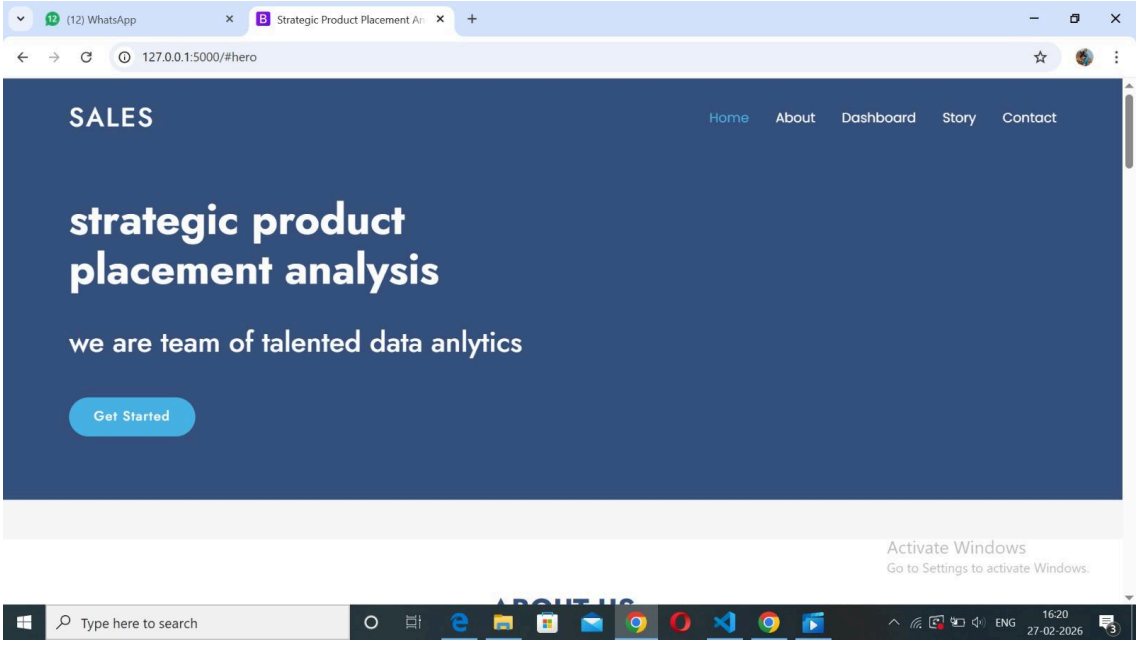
After signing into your public account the workbook is displayed. Now click on “settings” and then it will display to show sheets are disabled so enable it so, that all your sheets are visible.

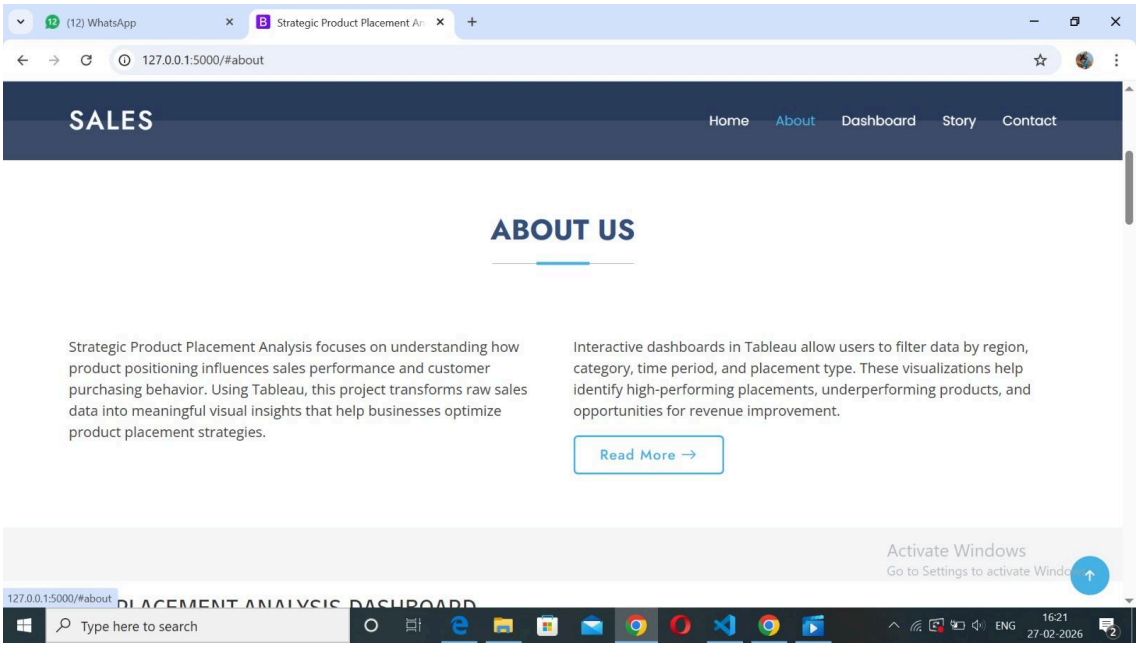
Now click on the dashboard sheet in the top right corner we have an option called share click on it then it will show like the above screenshot. Then, copy the embedded code and place the copied embedded code into your bootstrap template as shown in the reference video below.

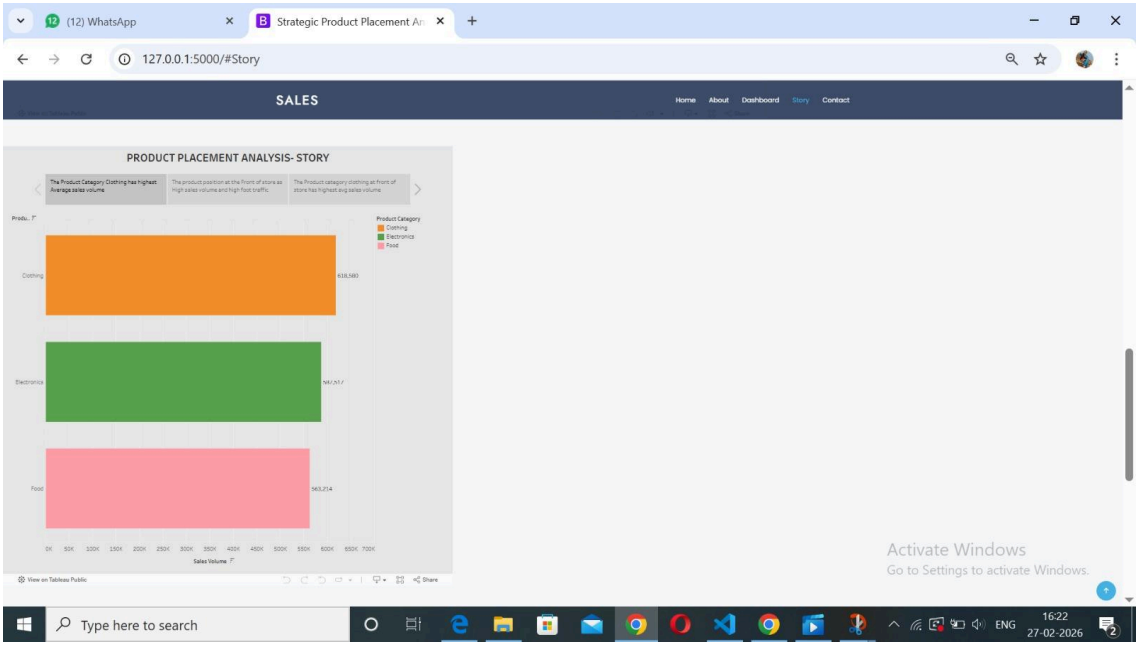


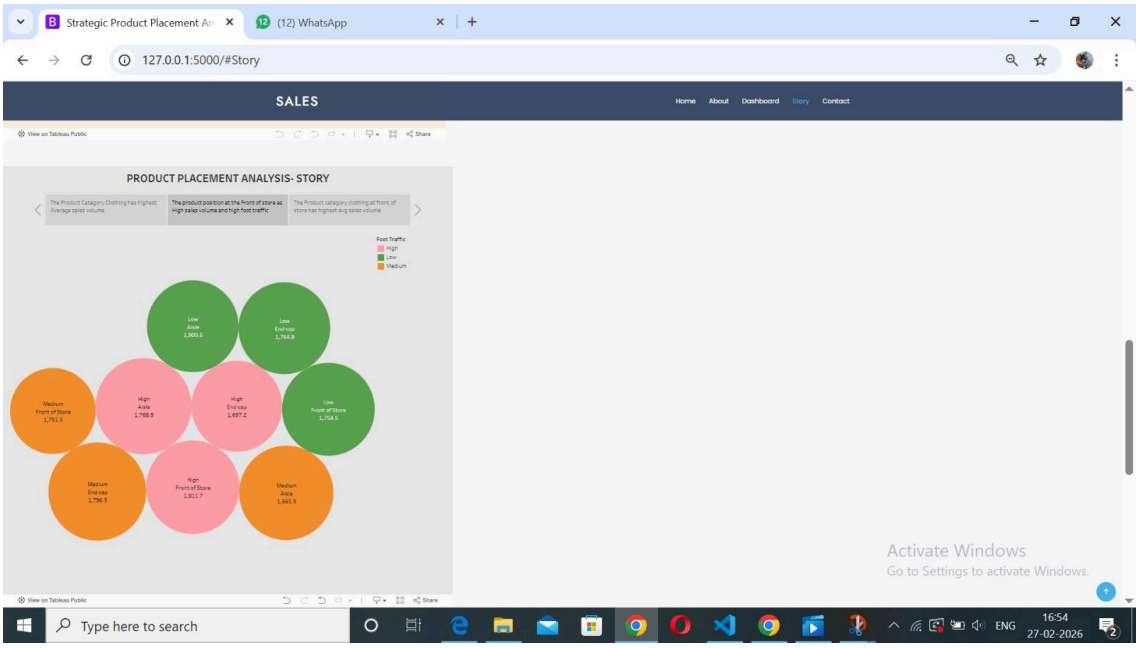
After copying the embedded code of the dashboard from Tableau Public and pasting it in the Bootstrap template then use the same procedure for the story also to copy the embedded code from the Tableau Public and paste it into the Bootstrap template.

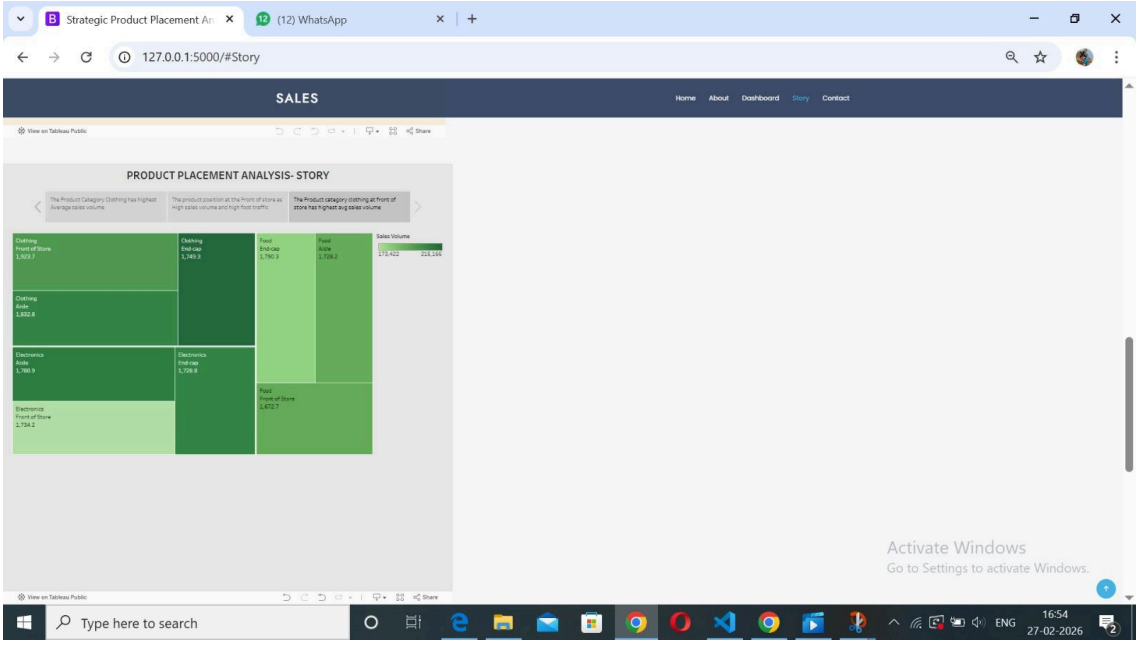
Dashboard and Story embed with UI With Flask

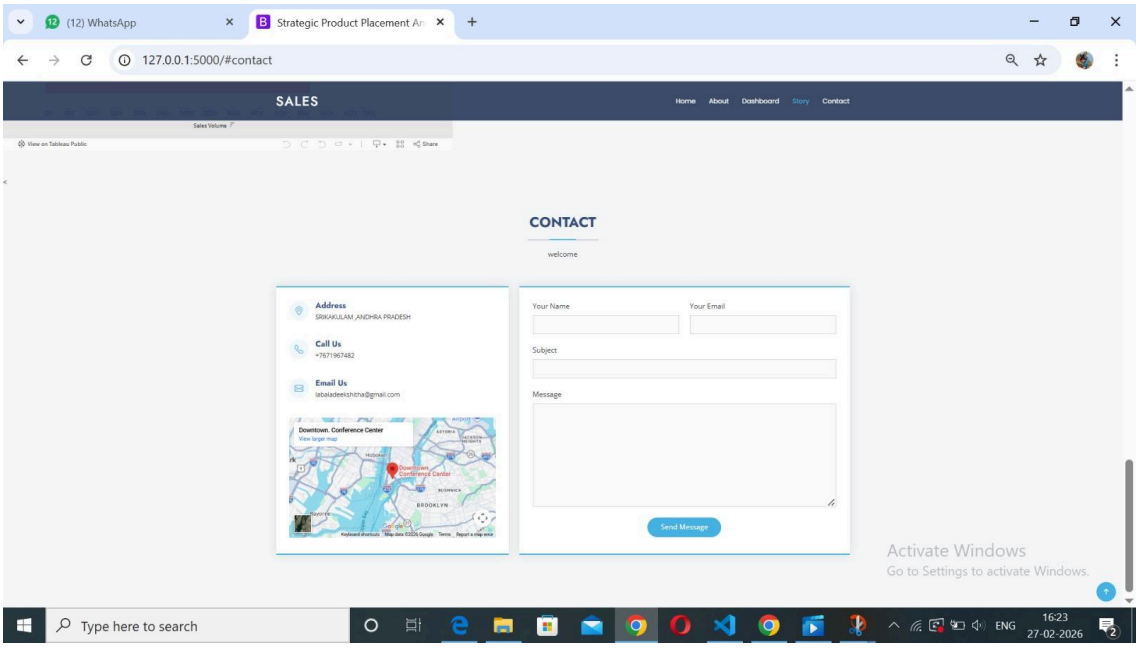

(12) WhatsApp
Strategic Product Placement An
127.0.0.1:5000/#hero
SALES
Home
About
Dashboard
Story
Contact
strategic product placement analysis
we are team of talented data anlytics
Get Started
Activate Windows
Go to Settings to activate Windows.
Type here to search
ENG
16:20
27-02-2026


(12) WhatsApp
Strategic Product Placement An
127.0.0.1:5000/#about
SALES
Home
About
Dashboard
Story
Contact
ABOUT US
Strategic Product Placement Analysis focuses on understanding how product positioning influences sales performance and customer purchasing behavior. Using Tableau, this project transforms raw sales data into meaningful visual insights that help businesses optimize product placement strategies.
Interactive dashboards in Tableau allow users to filter data by region, category, time period, and placement type. These visualizations help identify high-performing placements, underperforming products, and opportunities for revenue improvement.
Read More →
Activate Windows
127.0.0.1:5000/#about
Type here to search
ENG
16:21
27-02-2026


(12) WhatsApp
Strategic Product Placement An
127.0.0.1:5000/#Story
SALES
Home
About
Dashboard
Story
Contact
PRODUCT PLACEMENT ANALYSIS- STORY
The Product Category Clothing has highest Average sales volume
The product position at the Front of store as High sales volume and high foot traffic
The Product category clothing at front of store has highest avg sales volume
Product Category
Clothing
Electronics
Food
Clothing
Electronics
Food
618,580
947,517
563,214
0K 50K 100K 150K 200K 250K 300K 350K 400K 450K 500K 550K 600K 650K 700K
Sales Volume
View on Tableau Public
Share
Activate Windows
Go to Settings to activate Windows.
Type here to search
ENG
16:22
27-02-2026


Strategic Product Placement An
(12) WhatsApp
127.0.0.1:5000/#Story
SALES
Home
About
Dashboard
Story
Contact
View on Tableau Public
Share
PRODUCT PLACEMENT ANALYSIS- STORY
The Product Category Clothing has highest Average sales volume
The product position at the Front of store as High sales volume and high foot traffic
The Product category clothing at front of store has highest avg sales volume
Foot Traffic
High
Low
Medium
Low Aisle 1,900.5
Low End-cap 1,764.8
Medium Front of Store 1,751.5
High Aisle 1,768.9
High End-cap 1,697.2
Low Front of Store 1,754.5
Medium End-cap 1,796.5
High Front of Store 1,811.7
Medium Aisle 1,691.9
View on Tableau Public
Share
Activate Windows
Go to Settings to activate Windows.
Type here to search
ENG
16:54
27-02-2026





Project Demonstration & Documentation

A) Record an explanation Video for the project end to end solution

Creating a record explanation video for a project's end-to-end solution is crucial for ensuring clarity and transparency in its implementation.

B) Project Documentation-Step by step project development procedure

Create a document as per the template provided

Advantages

Data-Driven ROI: Replaces guesswork with exact "Sales Lift" metrics for every shelf position.
Visual Clarity: Uses Tableau heatmaps to instantly identify high-traffic "Hot Zones" and underperforming "Cold Zones".
Unified Insights: Merges siloed data from SQL databases and Google Drive into one accessible dashboard.
Real-Time Agility: Enables store managers to make immediate layout adjustments based on live performance data.
Consumer Understanding: Links demographics (Families, Seniors) to specific placement success for targeted marketing.

Disadvantages

Data Complexity: Requires significant effort in the "Data Preparation" phase to clean and transform raw POS records.
Integration Hurdles: Initial setup to connect Google Drive planograms with SQL databases can be technically challenging.
Maintenance Needs: Dashboards require regular updates to account for seasonal product changes and new SKUs.
High Initial Effort: Sprint-1 and Sprint-2 require heavy lifting from team members before any visual insights are produced.
Human Bias: Even with data, physical store constraints (e.g., fixed heavy shelving) may limit the ability to move products to "ideal" spots.

Conclusion

The Strategic Product Placement Analysis project successfully demonstrates how data-driven visualization can transform traditional retail management into a high-precision science. By bridging the gap between raw sales figures and physical store geography, the project provides a comprehensive solution for "unveiling" the hidden drivers of consumer behavior.

FUTURE SCOPE

Predictive AI Integration: Incorporating Machine Learning models to predict "Sales Lift" before a physical layout change is even made.

Real-Time IoT Sensors: Using in-store shelf sensors to update Tableau heatmaps instantly as products are picked up, rather than waiting for POS sync.

AR Visualization: Developing Augmented Reality (AR) tools for store managers to overlay Tableau performance metrics directly onto physical shelves using mobile devices.

Advanced Demographic Filtering: Integrating loyalty card data to track individual "Shopper Journeys" beyond broad categories like "Families" or "Seniors".

Automated Reordering: Linking shelf-level performance data to supply chain systems to automatically trigger restocks for high-velocity "Hot Zones".

Global Store Benchmarking: Expanding the Flask UI to compare placement effectiveness across multiple geographic locations or store formats.

APPENDIX

Source Code(if any)

Flask Folder link: